

CS Study Guide: Network Layer Core Concepts, Router Architecture, IPv4 Datagrams, and ICMP Protocols

Computer Networking Notes

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1 Core Functions of the Network Layer

The primary role of the Network Layer (OSI Layer 3) is to transport packets from a sending host to a receiving host across multiple intermediate networks and routers.

1.1 Routing vs. Forwarding

Routing (Control Plane): The network-wide process of determining the end-to-end paths that packets take from source to destination. Executes on a millisecond time scale via software algorithms (e.g., OSPF, BGP).

Forwarding (Data Plane): The local, router-specific action of transferring a packet from an input link interface to the appropriate output link interface. Executes on a nanosecond time scale via dedicated hardware circuits.

Real-World Analogy: Routing vs. Forwarding

Routing is like planning a cross-country road trip using a map before driving (calculating the optimal highways to take). **Forwarding** is like executing a single turn at an intersection based on street signs as you drive.

1.2 Packet Switching Paradigms

Feature	Datagram Switching	Virtual Circuit (VC) Switching
Connection Setup	None (Connectionless)	Required (Call Setup phase)
Router State	Stateless (No connection state)	Stateful (Stores VC table entries)
Packet Paths	Dynamic (Packets may take different paths)	Fixed (All packets follow reserved path)
Addressing	Full Destination IP in each packet	Short VC Identifier (VCI) in header
Examples	Standard Internet Protocol (IPv4/IPv6)	ATM, Frame Relay, MPLS

Table 1: Comparison of Datagram vs. Virtual Circuit Switching

2 Router Architecture

A high-performance router consists of two major operational planes: the **Control Plane** and the **Data Plane**.

2.1 Control Plane (Software Level)

- Executes routing algorithms (OSPF, BGP, RIP) on the Routing Processor.
- Computes the global **Routing Table**.
- Constructs the **Forwarding Information Base (FIB)** and pushes duplicate copies to each input line card for rapid local processing.

2.2 Data Plane (Hardware Level)

- **Input Ports:** Perform physical layer termination, data link layer decapsulation, lookup in local FIB memory, and packet queuing.
- **High-Speed Switching Fabric:** Connects input ports to output ports (e.g., via Shared Memory, Shared Bus, or Crossbar Matrix).
- **Output Ports:** Buffer outgoing packets, apply scheduling policies (e.g., FIFO, Priority, Weighted Fair Queuing), and handle data link layer encapsulation.

2.3 Forwarding Mechanisms

1. **Destination-Based Forwarding:** Forwarding decision relies solely on matching the destination IP address against the FIB using **Longest Prefix Matching (LPM)**.
2. **Generalized Forwarding (SDN):** Forwarding decision uses arbitrary header fields across Layers 2, 3, and 4 (e.g., OpenFlow match-action rules: forward, drop, modify, or clone).

2.4 Input Port Queuing & Buffer Overflow

Each interface possesses queue memory. If packets arrive at a rate faster than the switching fabric can transfer them, packets accumulate in the buffer:

- **Head-of-Line (HOL) Blocking:** A packet at the front of an input queue prevents trailing packets from moving forward, even if their target output ports are idle.
- **Packet Dropping:** When the queue memory buffer becomes completely full, incoming packets are dropped (tail-drop policy or Active Queue Management like RED).

3 IPv4 Datagram Header Structure

An IPv4 header has a minimum size of 20 bytes (without options) and a maximum size of 60 bytes.

Bits 0-3	Bits 4-7	Bits 8-15	Bits 16-31
Version (4 bits)	IHL (4 bits)	Type of Service / ToS (8 bits)	Total Length (16 bits)
Identification (16 bits)		Flags (3 bits)	Fragment Offset (13 bits)
Time to Live / TTL (8 bits)	Protocol (8 bits)	Header Checksum (16 bits)	
Source IP Address (32 bits)			
Destination IP Address (32 bits)			
Options (0-40 bytes, padded to 32-bit boundary)			

Table 2: IPv4 Header Format

3.1 Header Field Descriptions

- **Version (4 bits):** Specifies IP version (0100 for IPv4).
- **IHL / Internet Header Length (4 bits):** Measures header length in 32-bit (4-byte) words. Minimum value is $5 \times 4 = 20$ bytes; maximum value is $15 \times 4 = 60$ bytes.

- **Type of Service / ToS (8 bits):** Contains 6-bit DSCP for Quality of Service (QoS) dynamic priority and 2-bit ECN for explicit router congestion notification.
 - **Total Length (16 bits):** Total datagram size (header + data) in bytes. Maximum size is $2^{16} - 1 = 65,535$ bytes.
 - **TTL / Time to Live (8 bits):** Decrement by 1 at each router hop. Prevents packets from circulating indefinitely in routing loops. When $TTL = 0$, the packet is dropped and an ICMP Time Exceeded message is sent back.
 - **Protocol (8 bits):** Specifies upper-layer payload handler:
 - 1 = ICMP, 6 = TCP, 17 = UDP, 89 = OSPF.
 - **Header Checksum (16 bits):** Error detection over the IPv4 header fields only (recomputed at every hop due to TTL decrement).
 - **Padding:** Fills remaining space with zeros to ensure the header ends on a 32-bit (4-byte) alignment.
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4 IP Fragmentation and Reassembly

When an IP datagram's total size exceeds the link layer's **Maximum Transmission Unit (MTU)**, intermediate routers split the datagram into smaller fragments.

4.1 Key Rules of Fragmentation

1. **Reassembly Point:** Reassembly occurs **only at the destination host**, never at intermediate routers.
2. **Header Duplication:** Each fragment receives its own IP header.
3. **8-Byte Alignment Rule:** The payload length of every fragment (except the last fragment) **must be a multiple of 8 bytes**, because the 13-bit Fragment Offset field measures offset in 8-byte blocks.

4.2 Header Control Fields

- **Identification (16 bits):** Unique ID assigned to original datagram; shared by all fragments to aid reassembly.
- **DF (Don't Fragment - 1 bit):** If set to 1, routers must not fragment the packet. If the packet exceeds MTU, it is dropped and an ICMP error is returned.
- **MF (More Fragments - 1 bit):** Set to 1 for all fragments except the final fragment (0).
- **Fragment Offset (13 bits):** Indicates the starting position of the fragment's payload relative to the original unfragmented payload, expressed in 8-byte units:

$$\text{Fragment Offset} = \frac{\text{Data Payload Offset (in bytes)}}{8}$$

$$\text{Fragment Offset for } k\text{-th Fragment} = (k - 1) \times \frac{\text{Payload Size per Fragment}}{8}$$

4.3 Solved Mathematical Examples

Solved Math Problem 1 (From Slide 1)

Problem Statement:

A packet of size 2584 bytes including 32 bytes of header was fragmented for transmitting in a link with $\text{MTU} = X$ bytes. The 9th packet has a size of 272 bytes. It also has its MF bit set.

- I. Calculate the value of X .
- II. Calculate the fragment offset for the 7th packet.
- III. Find out the total number of fragmented packets.

Step-by-Step Solution:**I. Find MTU (X):**

Total Data Size of original packet = $2584 - 32 = 2552$ bytes.

The 9th fragment has total size = 272 bytes and MF = 1 (meaning it is a full non-final fragment).

Since full fragments fill the link capacity up to MTU:

$$\text{MTU } X = 272 \text{ bytes}$$

Data size per fragment = $272 - 32 = 240$ bytes.

Check alignment: $\frac{240}{8} = 30$ (valid integer).

II. Fragment Offset for 7th Packet:

Using formula with fragment index $k = 7$:

$$\text{Fragment Offset} = (7 - 1) \times \frac{240}{8} = 6 \times 30 = 180$$

III. Total Number of Fragmented Packets:

$$\text{Total Fragments} = \left\lceil \frac{\text{Total Data Size}}{\text{Data Size Per Fragment}} \right\rceil = \left\lceil \frac{2552}{240} \right\rceil = \lceil 10.633 \rceil = 11$$

Solved Math Problem 2 (From Slide 2)

Problem Statement:

A packet of size 7240 bytes including 40 bytes of header arrived at a router. The router can send at most 800 bytes at a time through the link ($\text{MTU} = 800$) and thus fragments accordingly.

- I. Calculate the number of fragments that will be created.
- II. Calculate the fragment size of the last fragment.
- III. Calculate the fragment offset of the 8th fragment.
- IV. Explain why the MF bit is always zero for the last fragment.

Step-by-Step Solution:

I. Number of Fragments Created:

Original Total Size = 7240 bytes, Header = 40 bytes.

Original Data Payload = $7240 - 40 = 7200$ bytes.

Maximum packet size per link = 800 bytes.

Max payload per fragment = $800 - 40 = 760$ bytes.

Check alignment: $\frac{760}{8} = 95$ (valid integer).

$$\text{Total Fragments} = \left\lceil \frac{7200}{760} \right\rceil = \lceil 9.473 \rceil = 10 \text{ fragments}$$

II. Fragment Size of Last Fragment:

Payload transmitted in first 9 fragments = $9 \times 760 = 6840$ bytes.

Payload remaining for 10th fragment = $7200 - 6840 = 360$ bytes.

Total size of 10th fragment (including header) = $360 + 40 = 400$ bytes.

III. Fragment Offset of 8th Fragment:

Using formula for $k = 8$:

$$\text{Fragment Offset} = (8 - 1) \times \frac{760}{8} = 7 \times 95 = 665$$

IV. Why MF Bit is 0 for the Last Fragment:

The **More Fragments (MF)** flag is a binary control field indicating whether additional fragments of the original datagram follow the current packet. $MF = 1$ means more fragments are incoming. For the final fragment, no further fragments exist, so $MF = 0$. This signals the destination host that it can complete reassembly once all preceding offsets are satisfied.

5 IP Addressing: Public vs. Private Addresses

5.1 Definitions

- **Public IP Addresses:** Globally unique addresses routable across the public Internet, assigned by IANA and Regional Internet Registries (RIRs).
- **Private IP Addresses:** Non-routable addresses reserved for local area networks (LANs) under RFC 1918. Routers on the public Internet drop packets with private destination IPs.

5.2 RFC 1918 Private IP Address Ranges

Real-World Analogy: Public vs. Private IPs

A **Public IP** is like your home's physical street address (globally unique; postal workers anywhere can deliver mail to it). A **Private IP** is like an apartment room number (Room 202 exists in thousands of buildings globally; meaningful only inside your specific building). **NAT (Network Address Translation)** acts as the building mailroom reception desk.

Class	IP Address Range	CIDR Prefix
Class A	10.0.0.0 – 10.255.255.255	10.0.0.0/8
Class B	172.16.0.0 – 172.31.255.255	172.16.0.0/12
Class C	192.168.0.0 – 192.168.255.255	192.168.0.0/16

Table 3: RFC 1918 Private Addressing Blocks

6 Internet Control Message Protocol (ICMP)

ICMP (Protocol Number 1) is used by network devices to send error messages and operational diagnostics.

6.1 Common ICMP Diagnostic Tools

- **Ping (Packet InterNet Groper):** Measures reachability and Round Trip Time (RTT). Sends ICMP Echo Request (Type 8, Code 0) and waits for ICMP Echo Reply (Type 0, Code 0).
- **Traceroute:** Identifies intermediate hop router IPs along a path. Sends packets with incremental TTL values ($TTL = 1, 2, 3 \dots$). When intermediate routers decrement TTL to 0, they return an ICMP Time Exceeded message (Type 11, Code 0), exposing their IP address.

6.2 Path MTU Discovery (PMTUD)

A process used by hosts to determine the minimum MTU along an end-to-end path without triggering IP fragmentation:

1. Source sends an IP packet with $DF = 1$ set in the header.
2. If a link along the route has an MTU smaller than the packet size, the router drops the packet.
3. The router sends an ICMP Destination Unreachable: Fragmentation Needed and DF Set message (Type 3, Code 4) back to the source, containing the router link MTU.
4. The source reduces its packet size to match the reported MTU and retransmits until packets reach the destination cleanly.

6.3 ICMP-Based Security Threats

- **ICMP Ping Flood (Zombie/Botnet Attack):** Attacker controls compromised hosts (zombies) to flood a target server with millions of ICMP Echo Requests, exhausting server CPU and bandwidth.
- **ICMP Amplification (Smurf Attack):** Attacker broadcasts ICMP Echo Requests with a **spoofed source IP** set to the victim's IP address across an amplifying subnetwork. Every host on that subnet replies to the victim simultaneously, multiplying attack bandwidth exponentially.